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CASE REPORT

Successful live birth after preimplantation genetic testing for combined nuclear *RNASEH2C* and mitochondrial *MT-TL1* variants

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Abstract

Introduction Preimplantation genetic testing (PGT) reduces reproductive risks for genetically high-risk families. To date, simultaneous PGT for nuclear and mitochondrial gene variants has rarely been reported. PGT targeting mitochondrial DNA (mtDNA) variants remains challenging and requires further clinical evidence.

Case Report We herein report a 3-year-old girl with severe global developmental delay. The proband harbored compound heterozygous *RNASEH2C* variants (maternal c.194G > A and paternal c.433C > T) and a maternally inherited m.3250 T > C variant in *MT-TL1* with 72.9% heteroplasmy. PGT was performed for her parents, including *RNASEH2C* haplotype phasing, m.3250 T > C variant load quantification, and chromosomal copy number variation analysis. All eight embryos exhibited lower m.3250 T > C heteroplasmy (0–17.8%) than the mother (20.2%–27.9%). One aneuploid embryo with 17.8% heteroplasmy was aliquoted into 16 specimens for technical validation, and consistent variant loads (18.2% ± 1.7%) were observed across all aliquots. An euploid embryo carrying heterozygous *RNASEH2C* c.194G > A and low-level m.3250 T > C heteroplasmy (1.8%) was transferred, resulting in the birth of a healthy boy with uneventful birth and two-year postnatal follow-up.

Conclusions This case represents a successful application of combined nuclear, mitochondrial, and chromosomal PGT in a family. Our findings provide valuable clinical evidence supporting blastocyst-stage PGT for mtDNA variants, indicating PGT to be a reliable reproductive option for females with low-level pathogenic mtDNA variants.

Keywords *RNASEH2C* · *MT-TL1* · Preimplantation genetic testing

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Introduction

Aicardi–Goutières syndrome (AGS) is a group of genetically heterogeneous disorders characterized by early-onset neurological regression and autoimmune abnormalities, predominantly impacting the brain, immune system, and skin [1]. Aicardi-Goutières syndrome type 3 (AGS3; MIM #610,329), resulting from biallelic mutations in the *RNASEH2C* gene, typically manifests in the neonatal period or early infancy with rapid clinical progression and leads to irreversible neurodevelopmental impairment in affected children [2].

Mitochondrial DNA (mtDNA) mutations lead to maternally inherited diseases. The coexistence of wild-type and mutant mtDNA results in heteroplasmy, and clinical disease commonly occurs when heteroplasmy exceeds a critical threshold [3]. *MT-TL1* mutations are associated with multiple conditions, including mitochondrial encephalomyopathy, lactic acidosis, and stroke-like episodes (MELAS), Leigh syndrome, mitochondrial cardiomyopathy, and maternally inherited diabetes and deafness, which severely compromise patients' quality of life and survival [4].

Preimplantation Genetic Testing (PGT) offers an effective approach for couples at risk of transmitting genetic disorders to have healthy children. It has been applied to more than 1,000 monogenic disorders following Mendelian inheritance patterns since 1990 [5, 6]. Due to the unique biological characteristics of mtDNA, the application of PGT for mtDNA disorders (PGT-MT) still faces multiple challenges. Offspring inherit only a small subset of maternal mtDNA with an unpredictable variant load due to the genetic bottleneck. The safe threshold of embryonic mtDNA heteroplasmy is mutation-specific [7]. Moreover, whether the mtDNA mutation level in a few trophectoderm (TE) cells is representative of the whole embryo remains a critical question. In 2012, an embryo with 12% m.3243A > G mutation load based on TE biopsy was transferred for the first time, resulting in a live birth with tissue-specific heteroplasmy ranging from 0 to 15% [8]. Another study observed a discrepancy of up to 22.7% in mutation load among different TE samples derived from the same embryo, indicating cellular heterogeneity [9]. Overall, more data are needed to provide evidence-based support for PGT-MT at the blastocyst stage.

This study reports a high-risk family harbouring variants in nuclear *RNASEH2C* and mitochondrial *MT-TL1*, associated with AGS3 and mitochondrial myopathy respectively. PGT for monogenic defects (PGT-M), PGT-MT, and PGT for aneuploidy (PGT-A) were simultaneously performed for this family, resulting in a healthy child. We observed changes in m.3250 T > C heteroplasmy during maternal inheritance and compared variant levels among different biopsied cells from the same embryo, providing a basis for the application of PGT-MT at the blastocyst stage.

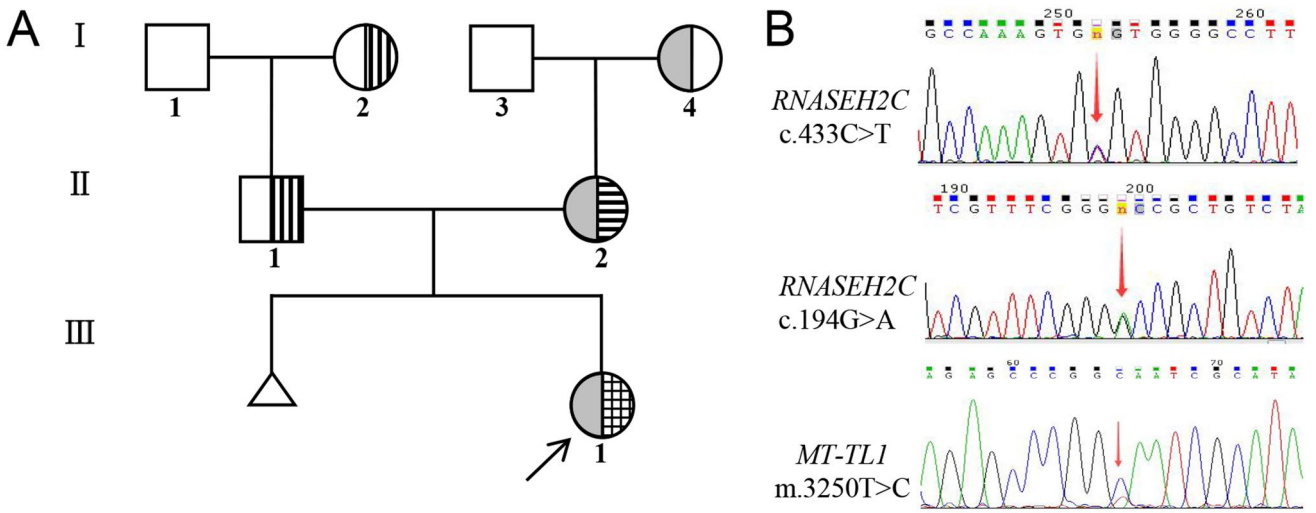


Fig. 1 Identification of the variants in *RNASEH2C* and *MT-TL1*. **A** Pedigree of the non-consanguineous Chinese family. Vertical stripes indicate the *RNASEH2C* gene (NM_032193.3) c.433C > T (p.Arg145Cys) variant, horizontal stripes indicate the c.194G > A (p.Gly65Asp) variant, and the grid pattern indicates individual carrying both variants. Gray background denotes the *MT-TL1* (NC_012920.1) m.3250 T > C mutation. The arrow indicates the proband. **B**. Sanger sequencing confirmation showing the nucleotide variants in the proband

Case report

Clinical presentation

The proband (Ⅲ1, Fig. 1) was a 3-year-old girl presenting with severe congenital global developmental delay. She had been unable to hold her head upright steadily, with absent visual tracking and impaired spontaneous grasping since birth. At 3 months of age, she developed progressive stimulus-induced startle spasms triggered by sound and touch, characterized by paroxysmal crying, limb hypertonia and rigid hand extension. Cranial CT performed at 10 months showed bilateral basal ganglia calcification, and brain MRI further confirmed cerebral atrophy. At age 3, her weight was 16 kg and head circumference 46 cm. She manifested typical chilblain-like cutaneous lesions on the auricles and fingers, together with severe psychomotor retardation, intellectual disability and dysphagia. No similar phenotypes were observed in other family members, indicating sporadic onset of the disorder in this family.

Genetic testing of the proband and family members

All tests were performed with the couple's informed consent. Trio whole-exome sequencing identified compound heterozygous variants in *RNASEH2C* (NM_032193.3) in the proband, a maternal c.194G > A (p.Gly65Asp) and a paternal c.433C > T (p.Arg145Cys). Sanger sequencing confirmed that c.433C > T was inherited from the paternal grandmother (Ⅰ2), while the origin of c.194G > A could not be determined due to the unavailability of the maternal grandfather's (Ⅰ3) sample (Table 1, Fig. 1, Supplementary Materials and Methods). Both variants were classified as likely pathogenic according to the ACMG guideline.

Meanwhile, the proband exhibited consistently high *MT-TL1* (NC_012920.1) m.3250 T > C heteroplasmy across multiple tissues, with mutation loads of 72.9% in peripheral blood, 72.7% in urine, and 76.4% in oral mucosa. Her mother presented low-level heteroplasmy (20.2%, 21.7%, and 27.9% in corresponding tissues, respectively). The maternal grandmother (Ⅰ4) carried an extremely low mutation load of 1.7% (Table 1, Supplementary Table 1). The *MT-TL1* m.3250 T > C variant impairs mitochondrial function by reducing ATP production and increasing reactive oxygen species (ROS) [10]. Although it has been associated with a spectrum of mitochondrial diseases, including exercise intolerance, fatigue, muscle weakness, and cardiomyopathy [11–13], the variant is annotated as a variant of uncertain significance (VUS) due to incomplete penetrance and likely benign in silico predictions [14, 15].

Genetic counseling before PGT

The *RNASEH2C* variants account for the proband's core phenotype, yet we cannot rule out a contributory role of the *MT-TL1* m.3250 T > C variant in her muscle weakness. The proband's parents (Ⅱ1 and Ⅱ2) requested PGT for both *RNASEH2C* and *MT-TL1* mutations to mitigate the reproductive risks.

In addition to counseling the couple on the general workflow, risks and limitations of routine PGT, we held targeted consultations focusing specifically on PGT-MT: (i) PGT-MT is indicated for female carriers with low heteroplasmic loads of pathogenic mtDNA variants. (ii) Sample heterogeneity. The biopsied cells may fail to accurately reflect the true variant load of the whole embryo. (iii) Embryo transfer. Embryos free of the known mtDNA variant shall be prioritized for transfer; embryos with variant loads below a safe threshold (e.g., 18%) are considered as the second option.

Table 1 The genotype information of families

Member ID	Gender	Age	Kinship	*RNASEH2C* genotype (NM_032193.3)	*MT-TL1* genotype (NC_012920.1)
Ⅰ1	Male	60	Proband's paternal grandfather	Normal	ND
Ⅰ2	Female	58	Proband's paternal grandmother	c.433C > T (p.Arg145Cys)	ND
Ⅰ4	Female	57	Proband's maternal grandmother	Normal	m.3250 T > C (B:1.7%)
Ⅱ1	Male	33	Proband's father	c.433C > T (p.Arg145Cys)	Normal
Ⅱ2	Female	33	Proband's mother	c.194G > A (p.Gly65Asp)	m.3250 T > C (B:20.2%;U:21.7%; O:27.9%)
Ⅲ1	Female	3	Proband	c.194G > A (p.Gly65Asp) c.433C > T (p.Arg145Cys)	m.3250 T > C (B:72.9%;U:72.7%; O:76.4%)

ND Not detected, *B* Blood, *U* Urine, *O* Oral mucosal cells

(iv) PGT-MT can only mitigate, rather than eliminate, the reproductive risk of affected offspring. The inheritance and segregation of mtDNA variants are highly dynamic. Although the variant load detected in trophectoderm biopsy may fall below the acceptable threshold, the heteroplasmic level can rise in specific tissues during embryonic development or postnatal life. Hence, the risk of disease in offspring cannot be completely ruled out. (v) Prenatal diagnosis and long-term postnatal follow-up are advised. Amniocentesis performed during the second trimester allows re-evaluation of the mtDNA variant load, and long-term clinical surveillance after birth facilitates monitoring dynamic shifts in heteroplasmy and associated disease risks.

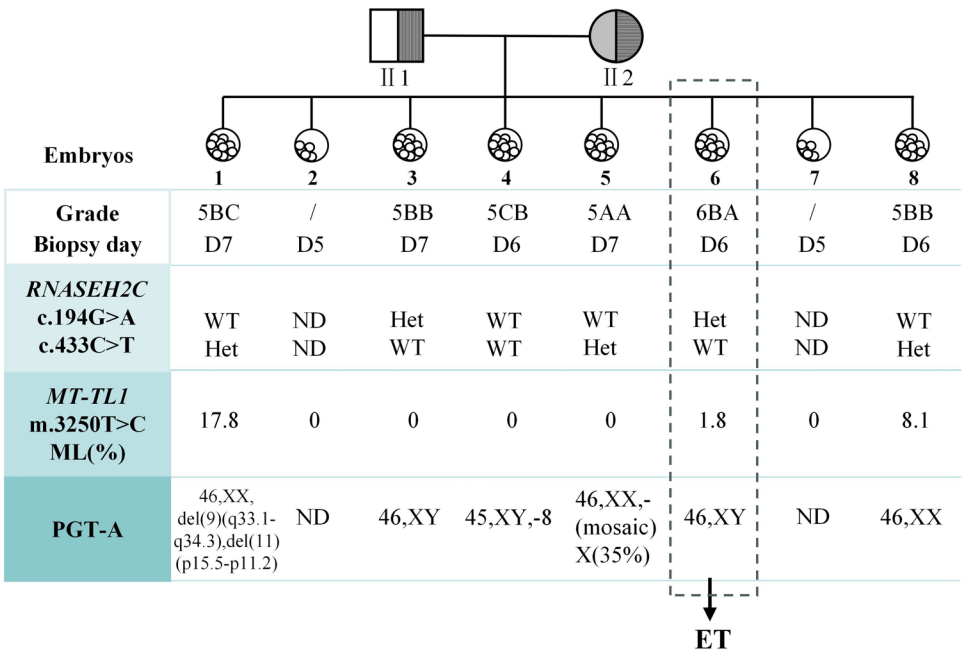
Pre-examination process of PGT

A customized multiplex PCR capture panel covering the *RNASEH2C* coding region and flanking 273 single nucleotide polymorphism (SNP) loci was designed to construct family haplotypes (Supplementary Materials and Methods), which enabled precise identification of parental variant origins in subsequent embryonic testing.

To validate the effect of single-cell whole-genome amplification (WGA) for mtDNA detection, single lymphocytes (3–5 cells/tube) were amplified using the REPLI-g Single Cell Kit (QIAGEN, Hilden, Germany) per the manufacturer's protocol, and the *MT-TL1* m.3250 T > C variant in the WGA products was detected by targeted PCR and next-generation sequencing (NGS) (Supplementary Materials and Methods). The mean mutation level was 71.8% ± 1.2% in the proband's lymphocytes and 20.2% ± 1.8% in her mother's lymphocytes, which was highly consistent with genomic DNA detection results (72.9% and 20.2%), indicating that WGA did not drastically alter mtDNA mutation load (Supplementary Table 1).

Intracytoplasmic sperm injection, embryo biopsy and testing

Ovarian stimulation, intracytoplasmic sperm injection and embryo biopsy were carried out as previously described [16]. A total of 8 oocytes were retrieved and successfully fertilized in the PGT cycle. Six zygotes developed to the blastocyst stage and underwent trophectoderm biopsy. *RNASEH2C* haplotype analysis revealed that one embryo (E4) was wild-type, two embryos (E3, E6) carried the maternal c.194G > A variant, and three embryos (E1, E5, E8) carried the paternal c.433C > T variant. NGS quantification showed undetectable *MT-TL1* m.3250 T > C mutation in E3, E4 and E5, a low load of 1.8% in E6, 8.1% in E8, and



Embryos	1	2	3	4	5	6	7	8
Grade Biopsy day	5BC D7	/ D5	5BB D7	5CB D6	5AA D7	6BA D6	/ D5	5BB D6
RNASEH2C c.194G>A c.433C>T	WT Het	ND ND	Het WT	WT WT	WT Het	Het WT	ND ND	WT Het
MT-TL1 m.3250T>C ML(%)	17.8	0	0	0	0	1.8	0	8.1
PGT-A	46,XX, del(9)(q33.1-q34.3),del(11)(p15.5-p11.2)	ND	46,XY	45,XY,-8	46,XX,-(mosaic) X(35%)	46,XY	ND	46,XX

Fig.2 Results of embryonic genetic testing. Vertical stripes indicate the *RNASEH2C* gene (NM_032193.3) c.433C > T variant, and horizontal stripes indicate the c.194G > A variant. Gray background denotes the *MT-TL1* (NC_012920.1) m.3250 T > C mutation. Blastocyst morphological grading was performed according to the Gardner scoring system. Each grade comprises three components: blastocoel expansion (1–6), inner cell mass (**A**-**C**), and trophectoderm (**A**-**C**). D5-D7 indicates the day of blastocyst development post-fertilization, as well as the day on which trophectoderm biopsy was performed; "/" denotes arrested embryos with no morphological scoring. WT: Wild-type; Het: Heterozygote; ND: Not detected; ML: Mutation load; ET: Embryo transfer; PGT-A: Preimplantation genetic testing for aneuploidy

17.8% in E1. Combined with PGT-A results, E1, E4, and E5 exhibited chromosomal copy number variations (CNVs), while E3, E6, and E8 were euploid. According to the results of PGT-M, PGT-MT and PGT-A, E3, E6 and E8 carried low disease risk and were eligible for embryo transfer (Fig. 2). Ultimately, E6 was prioritized for transfer due to its optimal Gardner morphological grade.

In addition, m.3250 T > C mutation was undetectable in the two arrested embryos (E2 and E7). E1 (17.8% heteroplasmy) was dissociated into single cells, and 16 samples (3–5 cells/tube) were collected (11 samples could not distinguish inner cell mass (ICM) from surrounding TE cells). The mutation loads ranged from 14.7% to 21.9%, with a mean of 18.2% ± 1.7% (95% CI: 17.3% to 19.1%, CV: 9.7%) (Table 2), which was consistent with the first biopsied TE results. These findings indicated that the mutation burden of a single TE biopsy was largely representative of the whole blastocyst.

Prenatal genetic diagnosis and postnatal follow-up

At 18 weeks of gestation, genetic analysis of amniotic fluid revealed a normal fetal karyotype. The fetus carried the heterozygous *RNASEH2C* c.194G > A variant, with an *MT-TL1* m.3250 T > C mutation load of 1.5%. A healthy boy was delivered at full term, and no m.3250 T > C variant was detected in his umbilical cord blood. At 2 years of age, re-examination of his peripheral blood and oral mucosal cells also showed no *MT-TL1* m.3250 T > C variant (Supplementary Table 1).

Discussion

The coexistence of both nuclear and mitochondrial gene mutations in a single individual is challenging for genetic counseling and reproductive intervention. This study describes a family harboring mutations in both the *RNASEH2C* and *MT-TL1* genes. Comprehensive NGS-based PGT was performed for the parents, resulting in the birth of a healthy child. Follow-up of this child was conducted from birth to 2 years of age, during which all developmental milestones were normal and no maternally derived mitochondrial mutations were detected. This study is the first to report the application of PGT-MT for the m.3250 T > C variant in *MT-TL1*, accumulating experience for the prevention of mitochondrial diseases associated with mtDNA mutations.

Identifying the underlying genetic cause is a prerequisite for PGT. Including this case, 12 AGS3 patients have been reported in China, among which the *RNASEH2C* c.194G > A variant is the most prevalent (10/12), suggesting a Chinese founder mutation effect, followed by the c.434G > T (p.Arg145Leu) variant (4/12) [2, 17, 18]. Herein, we identified a novel c.433C > T (p.Arg145Cys) variant at amino acid position 145, which causes replacement of a positively charged, hydrophilic arginine with an uncharged, less hydrophilic cysteine. Concurrently, the proband carries the *MT-TL1* m.3250 T > C variant at a high heteroplasmy level of 72.0%. Despite its classification as a VUS, we cannot exclude this variant as a contributor to

Table 2 Mutation load of m.3250 T > C in the *MT-TL1* gene in different biopsy samples of embryo 1

Sample No	Cell type	Cell number	Coverage depth (×)	Mutation load of 3250 T > C	Mean mutation load ± SD (95% CI, CV)
1	TE/ICM	3	93,314	18.1%	18.2% ± 1.7% (95% CI: 17.3% −19.1%, CV: 9.7%)
2	TE/ICM	3	92,970	17.7%	
3	TE/ICM	3	93,329	16.9%	
4	TE/ICM	3	92,752	16.3%	
5	TE/ICM	3	93,852	18.7%	
6	TE/ICM	5	95,633	19.7%	
7	TE/ICM	5	93,532	20.3%	
8	TE/ICM	5	95,184	17.5%	
9	TE/ICM	5	96,602	20.4%	
10	TE/ICM	5	97,201	21.9%	
11	TE/ICM	5	94,356	14.7%	
12	TE	3	94,815	18.6%	
13	TE	3	95,156	16.9%	
14	TE	3–5	94,114	18.5%	
15	TE	3–5	94,218	17.5%	
16	TE	5	93,083	17.5%	

TE Trophectoderm, *ICM* Inner cell mass, *SD* Standard deviation, *CI* Confidence interval, *CV* Coefficient of variation

the proband's muscle weakness. It may induce mitochondrial dysfunction and act as either a modifying factor or an independent pathogenic risk factor for the couple's future offspring. Therefore, both the nuclear *RNASEH2C* variants and mitochondrial *MT-TL1* variant are incorporated into the PGT panel to fully mitigate their reproductive risks.

Whether TE-derived mtDNA heteroplasmy accurately reflects the global embryonic status remains a key concern. Treff et al. confirmed negligible variations in m.3243A > G mutation loads across blastocyst cell clusters (SD: 2.1%–5.0%), and reported a strong correlation between TE and ICM values ($r^2 = 0.97$) [8]. This view was further supported by Rubens et al., who additionally compared hypervariable region 2 (HV2) heteroplasmy levels among single TE cells. No significant single-cell differences were observed in highly heteroplasmic embryos (> 80% heteroplasmy), whereas marked differences were detected in one embryo with moderate heteroplasmy of around 50%. Accordingly, sampling 5–10 TE cells were recommended in PGT-MT to reduce potential inter-cell variability [19]. Clinically, we routinely biopsy 3–5 TE cells for PGT. One aneuploid blastocyst with a 17.8% mutation load was divided into multiple subgroups with 3–5 cells each. The low coefficient of variation (9.7%) in mtDNA mutation levels across different subgroups demonstrated that sampling 3–5 TE cells enables stable and representative assessment of embryonic-wide mtDNA mutation load. Consistently with these observations, Mertens et al. proposed that mtDNA mutations originating in primordial germ cells and surviving the germ-line genetic bottleneck are evenly distributed in oocytes and segregate equally into daughter cells during early embryonic development [20].

PGT-MT is suitable for women with low mtDNA mutation loads, who have a high probability of obtaining usable embryos below the safe threshold [21]. A meta-analysis of 159 heteroplasmic mtDNA mutations across 327 families (excluding m.3243A > G, m.8344A > G, and m.8993 T > C/G) showed that individuals with a muscular heteroplasmy level ≤ 18% had a ≥ 95% probability of being unaffected [22]. Using this as a reference and combining the mutation levels in various tissues of the asymptomatic mothers in this study (20.2%−27.9%), we proposed 18% as the safety threshold for the m.3250 T > C mutation. Notably, the mutation loads of all embryos in this PGT cycle were below 18%, and nearly two-thirds (5/8) carried no detectable mutation. The mutation load in embryos showed a downward trend compared with maternal somatic cells, suggesting that m.3250 T > C may undergo negative selection during oogenesis in this individual. A similar finding has been documented in PGT for a female with m.3243A > G heteroplasmy ranging from 7.3% to 28%, where most of her embryos were mutation-free [23]. In addition to random genetic drift, other mechanisms influence the transmission of pathogenic mtDNA. Floros et al. proposed that metabolic shift from glycolysis to oxidative phosphorylation occurs during the development of embryonic primordial germ cells, which exposes deleterious mtDNA variants to selective pressure and mediates their preferential elimination [24].

To date, more than 100 PGT-MT cycles have been performed worldwide[22, 25]. Notably, Mitalipov et al. reported a child born after PGT-MT who developed a complex multisystem disorder. TE biopsy revealed 12% m.3243A > G heteroplasmy, while postnatal mutation loads in blood and urine exceeded 40% at both 6 weeks and 18 months of age [26]. This discrepancy may stem from limited PGT detection precision, potential intercellular heterogeneity, or post-birth heteroplasmy shifts. It should be emphasized that PGT-MT serves only as a risk-reduction strategy instead of a definitive diagnostic assay. Distinct from nuclear variants, mtDNA demonstrates highly dynamic inheritance and segregation. Low variant loads observed in TE biopsy cannot rule out subsequent elevation of mutational burden in specific tissues, and complete absence of pathogenic mtDNA in the offspring cannot be guaranteed. Considering these special features of mitochondrial genetics, patients must be fully informed of PGT-MT limitations in genetic counseling. Long-term follow-up of children conceived via PGT-MT is therefore essential to assess the safety of this technique and advance our understanding of mtDNA inheritance [27].

In conclusion, this study successfully performs comprehensive PGT, including haplotype phasing, quantitative detection of mtDNA mutation load, and chromosomal CNV analysis, in a family concurrently carrying mutations in the nuclear gene *RNASEH2C* and the mitochondrial gene *MT-TL1*. This study validates the effectiveness of PGT in reducing the genetic risk of complex diseases and provides evidence-based support for the application of PGT-MT at the blastocyst stage.

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Author contribution Conceptualization: Guangxiu Lu, Yue-Qiu Tan, Juan Du, Ge Lin; experimental operation: Xiao Hu, Yi Zhang, Zhenxing Wan, Jing Dai, Yifan Gu, Lu Zhou; funding acquisition: Xiao Hu, Ge Lin; clinical resources: Fei Gong, Guangxiu Lu; project administration and supervision: Yue-Qiu Tan, Juan Du; writing—original draft: Xiao Hu; writing—review and editing: Juan Du, Ge Lin. All authors reviewed the manuscript.

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Data availability All raw data in this study have not been deposited in a public repository due to privacy and ethical restrictions, but are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate This study was conducted in accordance with the Declaration of Helsinki and was approved by the Ethics Committee of the Reproductive & Genetic Hospital of CITIC-Xiangya (LL-SC-2024-031-01). Written informed consent was obtained from the participating couple.

Consent to participate Written informed consent was obtained from the parents.

Consent for publication Written informed consent was obtained from the participants for the publication of this study and all accompanying images and data.

Competing interests The authors declare no competing interests.

References

1. Rice G, Patrick T, Parmar R, Taylor CF, Aeby A, Aicardi J, et al. Clinical and molecular phenotype of Aicardi-Goutières syndrome. Am J Hum Genet. 2007;81:713–25. https://doi.org/10.1086/521373.
2. Wang W, Wang W, He TY, Zou LP, Li WD, Yu ZX, et al. Analysis of clinical characteristics of children with Aicardi-Goutières syndrome in China. World J Pediatr. 2022;18:490–7. https://doi.org/10.1007/s12519-022-00545-1.
3. Glynos A, Bozhilova LV, Frison M, Burr S, Stewart JB, Chinnery PF. High-throughput single-cell analysis reveals progressive mitochondrial DNA mosaicism throughout life. Sci Adv. 2023;9:eadi4038. https://doi.org/10.1126/sciadv.adi4038.
4. Gorman GS, Chinnery PF, DiMauro S, Hirano M, Koga Y, McFarland R, et al. Mitochondrial diseases. Nat Rev Dis Primers. 2016;2:16080. https://doi.org/10.1038/nrdp.2016.80.
5. Handyside AH, Kontogianni EH, Hardy K, Winston RM. Pregnancies from biopsied human preimplantation embryos sexed by Y-specific DNA amplification. Nature. 1990;344:768–70. https://doi.org/10.1038/344768a0.
6. Schadwell A, Whiting O, Xanthopoulou L, Colls P, Bakosi E, Goodall NN, et al. Universal preimplantation genetic testing for monogenic disease (Karyomapping): diagnosis of >1000 unique disorders with no detected misdiagnoses. Hum Reprod. 2025;40:2440–8. https://doi.org/10.1093/humrep/deaf198.
7. Sallevelt SC, Dreesen JC, Drüsedau M, Spierts S, Coonen E, van Tienen FH, et al. Preimplantation genetic diagnosis in mitochondrial DNA disorders: challenge and success. J Med Genet. 2013;50:125–32. https://doi.org/10.1136/jmedgenet-2012-101172.
8. Treff NR, Campos J, Tao X, Levy B, Ferry KM, Scott RT Jr. Blastocyst preimplantation genetic diagnosis (PGD) of a mitochondrial DNA disorder. Fertil Steril. 2012;98:1236–40. https://doi.org/10.1016/j.fertnstert.2012.07.1119.
9. Heindryckx B, Neupane J, Vandewoestyne M, Christodoulou C, Jackers Y, Gerris J, et al. Mutation-free baby born from a mitochondrial encephalopathy, lactic acidosis and stroke-like syndrome carrier after blastocyst trophectoderm preimplantation genetic diagnosis. Mitochondrion. 2014;18:12–7. https://doi.org/10.1016/j.mito.2014.08.005.
10. Campbell T, Lou X, Slone J, Brown J, Bromwell M, Liu J, et al. Mitochondrial genome variant m.3250T>C as a possible risk factor for mitochondrial cardiomyopathy. Hum Mutat. 2021;42:177–88. https://doi.org/10.1002/humu.24143.
11. Ogle RF, Christodoulou J, Fagan E, Blok RB, Kirby DM, Seller KL, et al. Mitochondrial myopathy with tRNA(Leu(UUR)) mutation and complex I deficiency responsive to riboflavin. J Pediatr. 1997;130:138–45. https://doi.org/10.1016/s0022-3476(97)70323-8.
12. Darin N, Hedberg-Oldfors C, Kroksmark AK, Moslemi AR, Kollberg G, Oldfors A. Benign mitochondrial myopathy with exercise intolerance in a large multigeneration family due to a homoplasmic m.3250T>C mutation in MTTL1. Eur J Neurol. 2017;24:587–93. https://doi.org/10.1111/ene.13249.
13. Lu Y, Zhao D, Yao S, Wu S, Hong D, Wang Q, et al. Mitochondrial tRNA genes are hotspots for mutations in a cohort of patients with exercise intolerance and mitochondrial myopathy. J Neurol Sci. 2017;379:137–43. https://doi.org/10.1016/j.jns.2017.05.056.
14. Lott MT, Leipzig JN, Derbeneva O, Xie HM, Chalkia D, Sarmady M, et al. mtDNA Variation and Analysis Using Mitomap and Mitomaster. Curr Protoc Bioinformatics. 2013;44:1.23.1–1.23.26. https://doi.org/10.1002/0471250953.bi0123s44.
15. Wong LC, Chen T, Wang J, Tang S, Schmitt ES, Landsverk M, et al. Interpretation of mitochondrial tRNA variants. Genet Med. 2020;22:917–26. https://doi.org/10.1038/s41436-019-0746-0.
16. Zhou S, Xie P, Zhang S, Hu L, Luo K, Gong F, et al. Complex mosaic blastocysts after preimplantation genetic testing: prevalence and outcomes after re-biopsy and re-vitrification. Reprod Biomed Online. 2021;43:215–22. https://doi.org/10.1016/j.rbmo.2021.04.006.
17. Wu J, Lu AD, Zhang LP, Zuo YX, Jia YP. Study of clinical outcome and prognosis in pediatric core binding factor-acute myeloid leukemia. Zhong hua Xue Ye Xue Za Zhi. 2019;40:52–7. https://doi.org/10.3760/cma.j.issn.0253-2727.2019.01.010.
18. Wang Q, Han Y, Zhou X, Cheng S, Wang X, Chen X, et al. RNASEH2C c.194G>A is a Chinese-specific founder mutation in three unrelated patients with Aicardi-Goutières syndrome 3. Clin Genet. 2023;104(2):259–65. https://doi.org/10.1111/cge.14343.
19. Rubens P, Mayeur A, Chatzovoulou K, Gigarel N, Monnot S, Rötig A, et al. Profiling mitochondrial DNA variant segregation during human preimplantation development: a prerequisite to preimplantation genetic testing for mitochondrial DNA-related disorders. Hum Reprod. 2025;40:956–61. https://doi.org/10.1093/humrep/deaf050.
20. Mertens J, Regin M, De Munck N, Couvreu de Deckersberg E, Belva F, Sermon K, et al. Mitochondrial DNA variants segregate during human preimplantation development into genetically different cell lineages that are maintained postnatally. Hum Mol Genet. 2022;31:3629–42. https://doi.org/10.1093/hmg/ddac059.
21. Hyslop LA, Blakely EL, Aushev M, Marley J, Takeda Y, Pyle A, et al. Mitochondrial donation and preimplantation genetic testing for mtDNA disease. N Engl J Med. 2025;393:438–49. https://doi.org/10.1056/NEJMoa2415539.
22. Hellebrekers DM, Wolfe R, Hendrickx AT, de Coo IF, de Die CE, Geraedts JP, et al. PGD and heteroplasmic mitochondrial DNA point mutations: a systematic review estimating the chance of healthy offspring. Hum Reprod Update. 2012;18:341–9. https://doi.org/10.1093/humupd/dms.
23. Spath K, Babariya D, Konstantinidis M, Lowndes J, Child T, Grifo JA, et al. Clinical application of sequencing-based methods for parallel preimplantation genetic testing for mitochondrial DNA disease and aneuploidy. Fertil Steril. 2021;115:1521–32. https://doi.org/10.1016/j.fertnstert.2021.01.026.
24. Floros VI, Pyle A, Dietmann S, Wei W, Tang WCW, Irie N, et al. Segregation of mitochondrial DNA heteroplasmy through a developmental genetic bottleneck in human embryos.

Nat Cell Biol. 2018;20:144–51. https://doi.org/10.1038/s41556-017-0017-8.
25. Smeets HJM, Sallevelt SCEH, Herbert M. Reproductive options in mitochondrial disease. Handb Clin Neurol. 2023;194:207–28. https://doi.org/10.1016/B978-0-12-821751-1.00004-X.
26. Mitalipov S, Amato P, Parry S, Falk MJ. Limitations of preimplantation genetic diagnosis for mitochondrial DNA diseases. Cell Rep. 2014;7:935–7. https://doi.org/10.1016/j.celrep.2014.05.004.
27. Li D, Burgstaller JP, Poulton J. How mitochondrial DNA complicates pre-implantation genetic screening and management. Expert Rev Mol Diagn. 2025;25:637–40. https://doi.org/10.1080/14737159.2025.2545967.

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